

Machine Learning 1

Lecture 1

Course Introduction & Probability Basics

Alejandro Monroy

Learning goals

- Define Machine Learning and the three main paradigms of supervised, unsupervised and reinforcement learning
- Describe and use the properties of probability calculus and random variables
- Compute marginal, conditional and posterior distributions and assess random variable independence
- Compute and interpret random variable moments, specifically the expected value, variance and covariance

Lecture 1

Course Introduction & Probability Basics

Alejandro Monroy

Outline

Course introduction	2
Admin	12
Probability theory	18

Lecture 1

Course Introduction & Probability Basics

Alejandro Monroy

Hi!



Alex Monroy
Junior Lecturer @ GSI

We will meet in many MSc AI courses!

- Machine Learning 1
- Deep Learning 1
- FACT
- Reinforcement Learning
- Deep Learning 2
- Foundation Models
- Recommender Systems

What is Machine Learning?

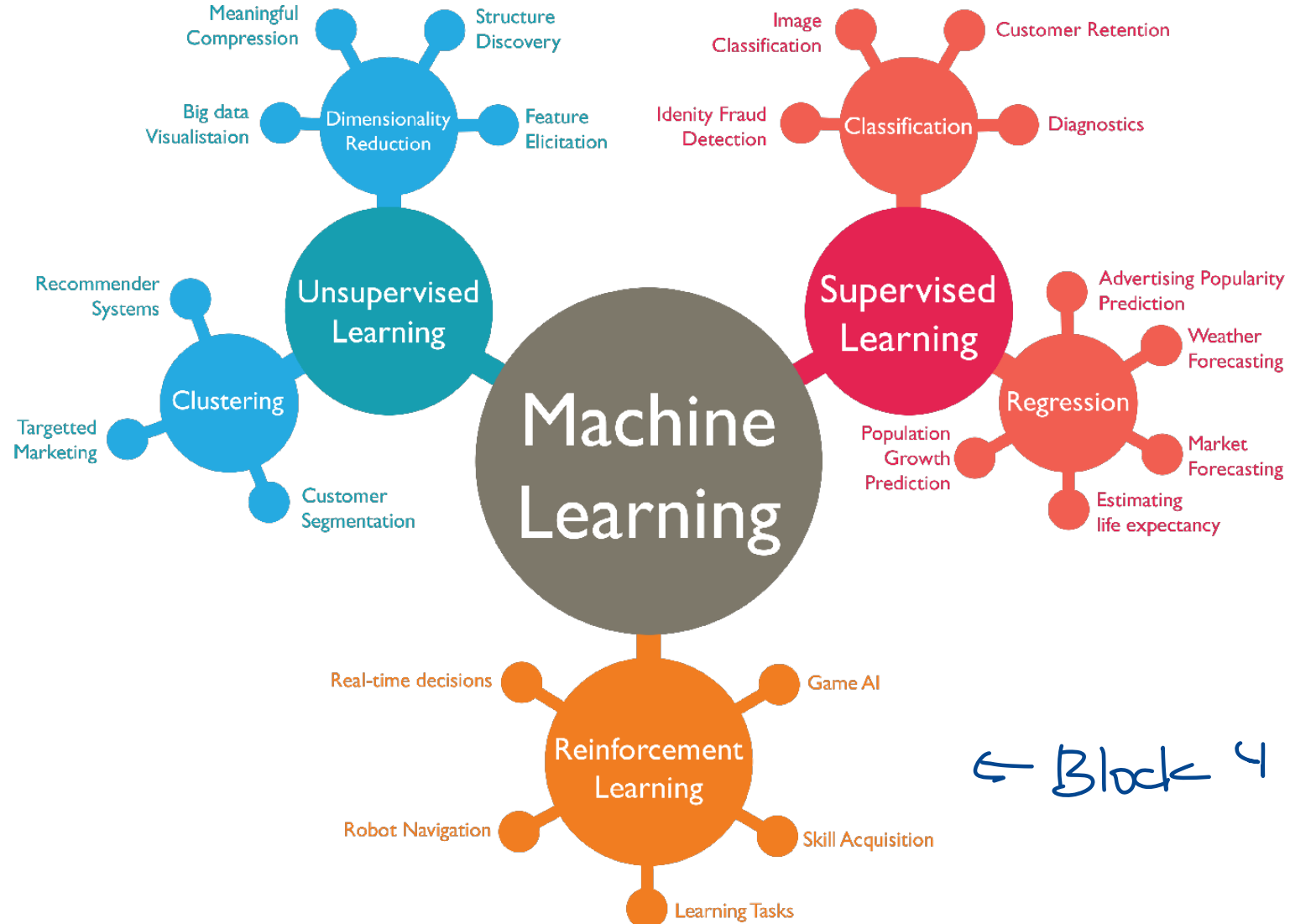
Artificial Intelligence

Programs that can use information to make decisions or predictions without active human involvement

Machine Learning

Models that aren't explicitly programmed, but **learn** patterns from **experience** to make **predictions** about new data

The main ML tasks



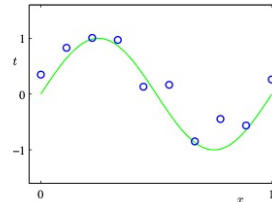
← Block 4

The main ML tasks – Supervised Learning

Task: Find a function f such that $f(\mathbf{x}) \approx t$ for all known and unknown pairs $(\mathbf{x}, t)$.

input features → target

Dataset



features: $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$

$\mathbf{x} =$

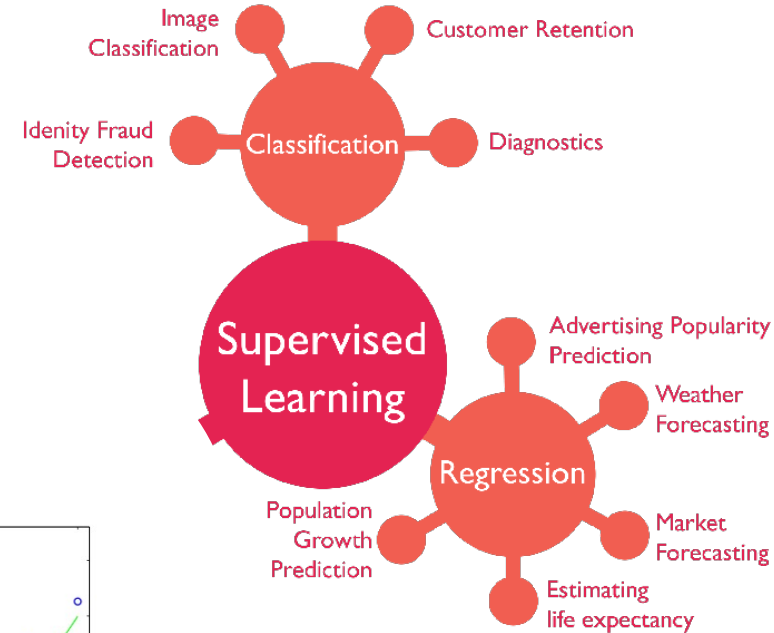
$x = 0.25$

targets: $\{t_1, \dots, t_N\}$

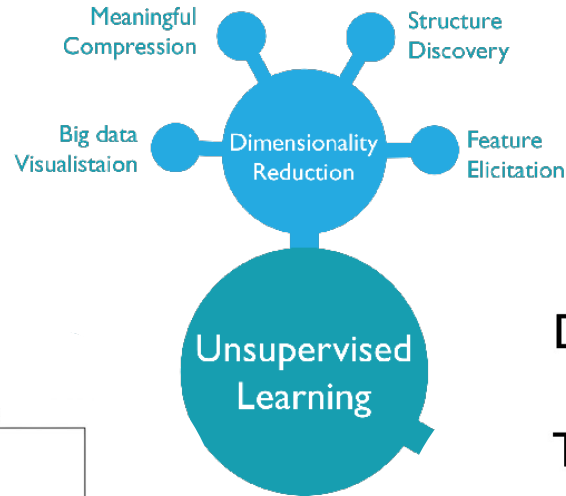
$t = 2$

$t = 0.707$

CLASSIFICATION REGRESSION



The main ML tasks – Unsupervised Learning



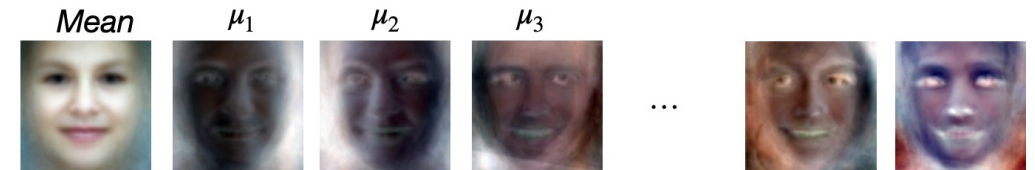
Task: Reduce the number of dimensions of $\mathbf{x}$

Dataset:

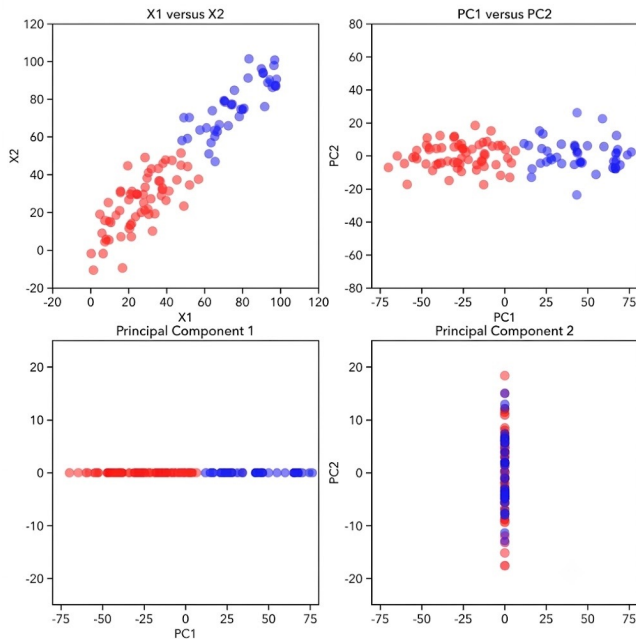


Task: Compress image

Method: Expand along principle components (PCA)

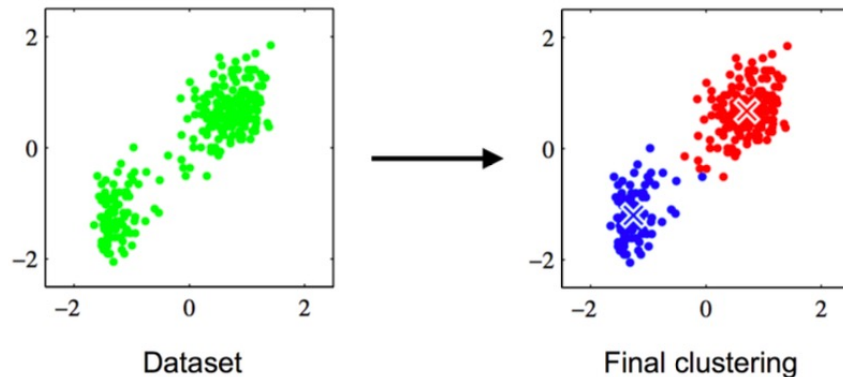
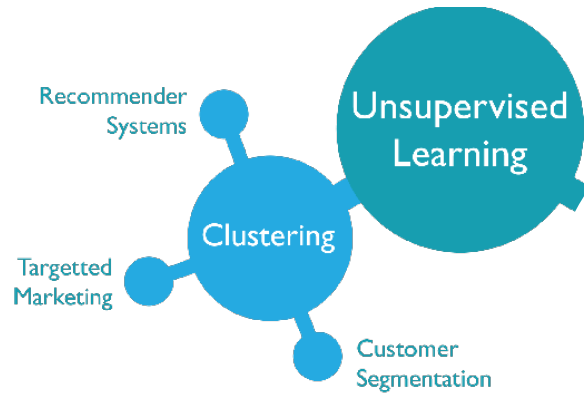
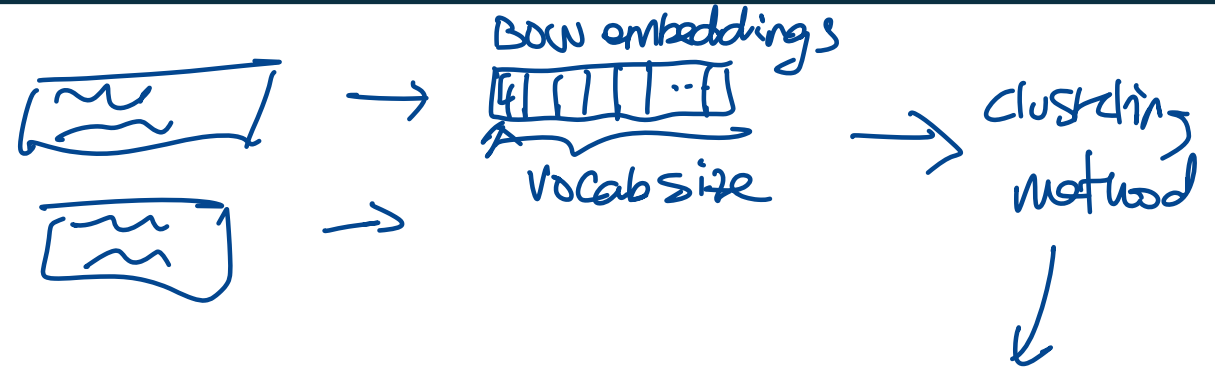


Result:



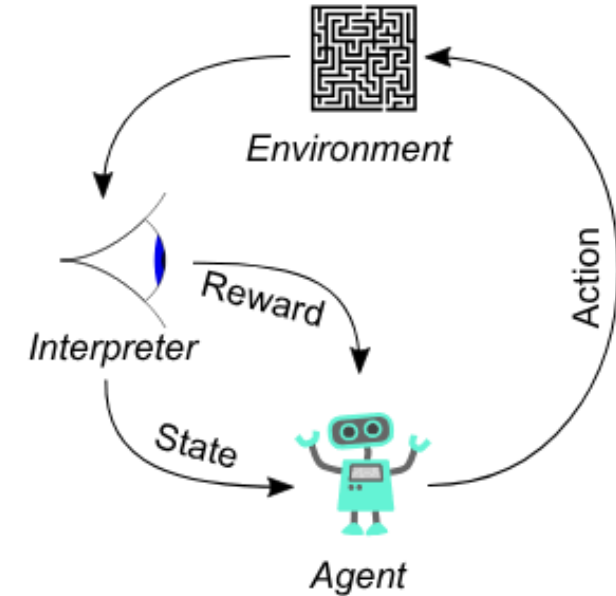
The main ML tasks – Unsupervised Learning

Task: Assign every datapoint x to a cluster.



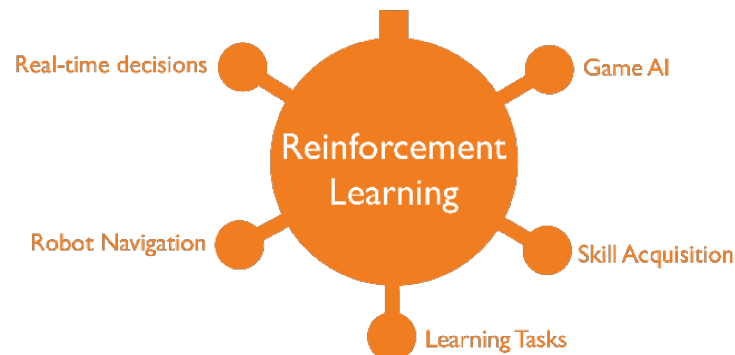
The main ML tasks – Reinforcement learning

Task: Find actions to take in a given state in order to maximize a reward.

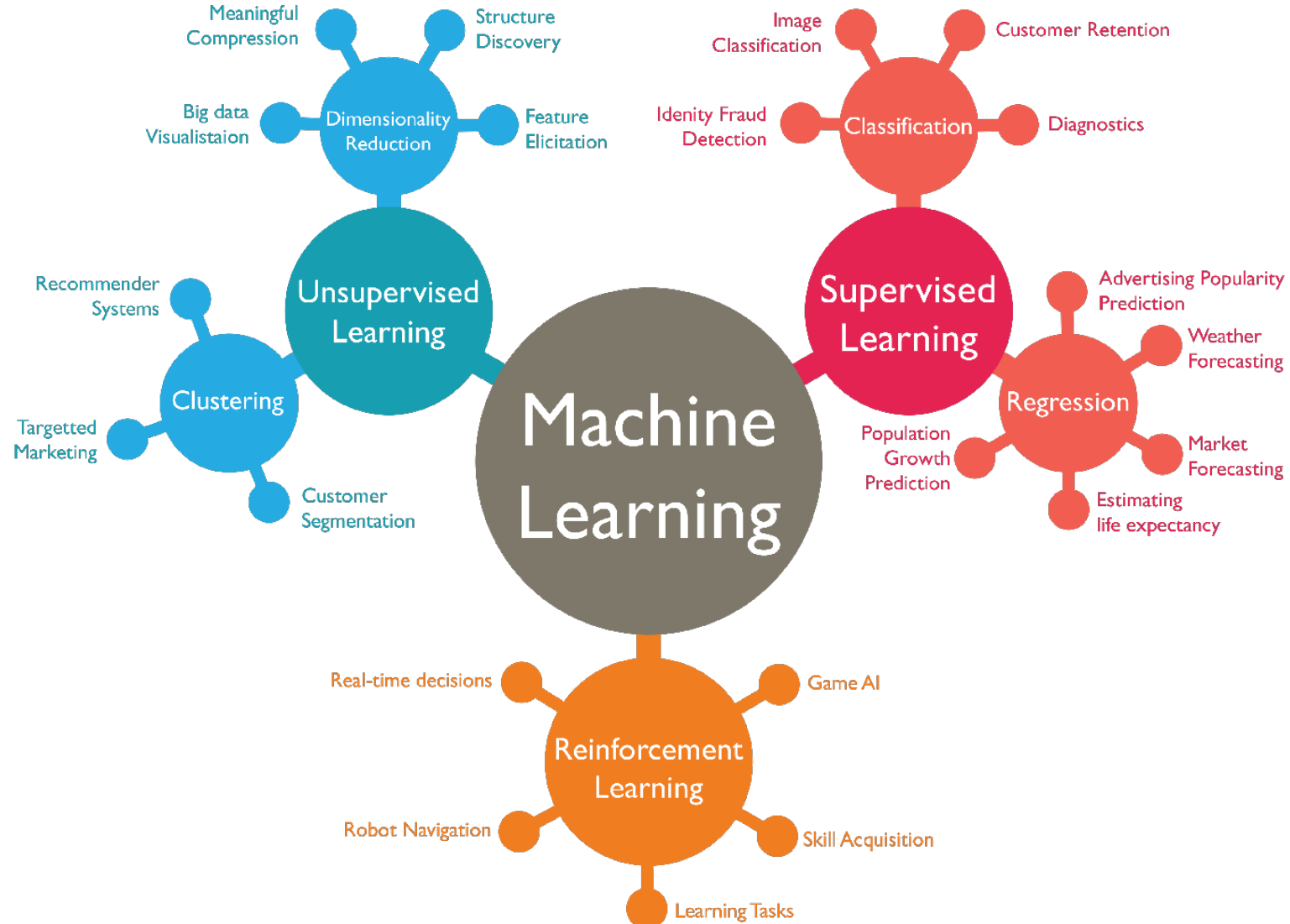


Source: Wikipedia

**You will learn more
in the RL course!
(block 4)**

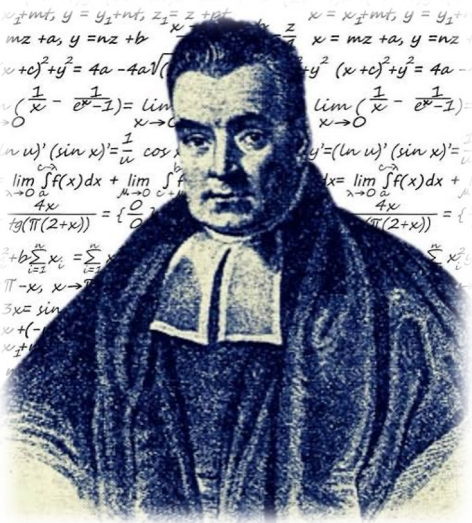


The main ML tasks



Course outline

The probabilistic view



W1

Probabilistic modeling

Random variables, distributions, maximum likelihood estimation, MAP

W7

Gaussian Processes

W6

Kernel Methods

W5

Latent variable models

W4

Neural Networks

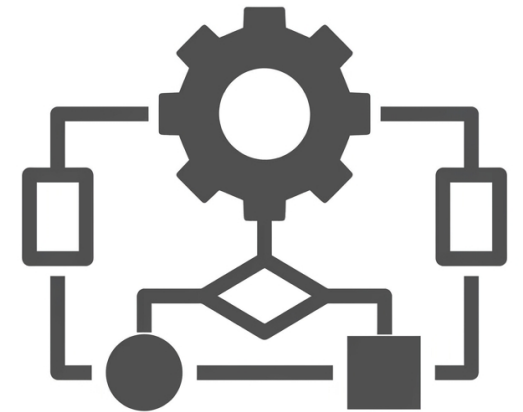
W3

Classification

W2

Regression

The algorithmic view



Optimization

Exact/analytic, (stochastic gradient descent, constrained optimization, 2nd order)

Mathematical tools

Multi-variate calculus, linear algebra

After completing this course, you will...

- Understand basic machine learning models such as Logistic Regression, Mixture of Experts, and Support Vector Machines.
- Master different estimation and optimization techniques, especially **Maximum Likelihood Estimation**, which form the basis of most ML methods.
- Have the mathematical tools required to understand and work with state-of-the-art machine learning methods.

Admin

The teaching team



Erik Bekkers

Associate professor @ AMLAB
Course coordinator / Lecturer



Alex Monroy

Junior Lecturer @ GSI
Head TA / Lecturer



Floor Eijkelboom

PhD Candidate @ AMLab
Math4AI Lecturer

Teaching assistants

Group 1, 2, 3



Daniel
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Vergnano

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Graff

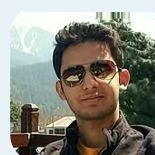


Hana
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Alex
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Group 7, 8, 9



Agneya
Bhardwaj

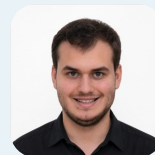


Naval
Bhagat



Xanthos
Kekkios

Group 10, 11, 12



Andrei
Nicula



Francesco
Massafra



Mathan
Sundarrajan

Learning activities

Every week:

- 2 lectures (Monday & Wednesday)
- 1 Math4AI lecture (Tuesday)
- 2 practical sessions (Wednesday/Thursday & Friday)

Exams:

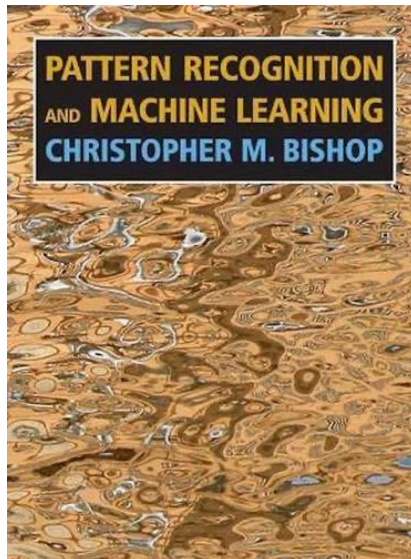
- HW Exam 1: Thursday September 17 (Week 3)
 - HW Exam 2: Thursday October 8 (Week 6)
 - Final Exam: Friday October 23 (Week 8)
 - Resit Exam: Friday January 8
- }  **Not in Science Park!**

Schedule and rooms can be found on [DataNose](#)

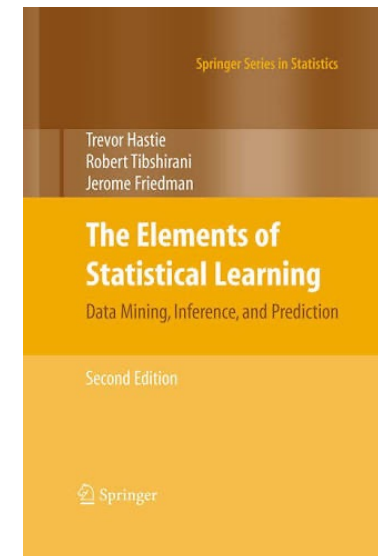
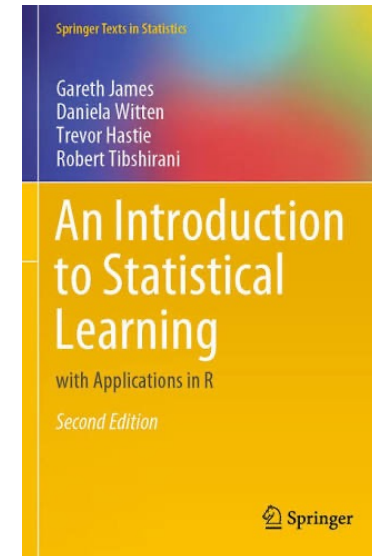
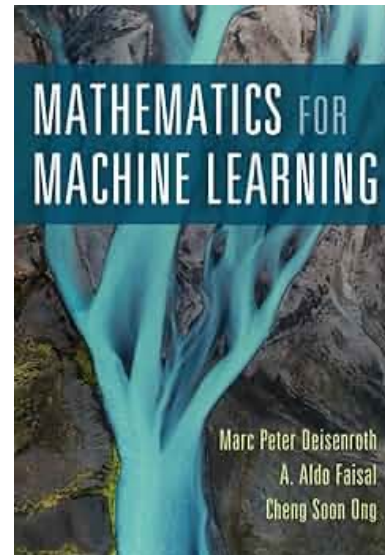
Learning activities - Lectures

- Lectures will ***not be recorded***
- Slides will be posted on Canvas
- For reference/preparation see videos and lecture notes on the [course website](#)

Main book



Other useful books



Learning activities – Math4AI

- Go over the mathematical prerequisites for the course
- Every week we will focus on hard math concepts that will appear in that week's lectures
- <https://www.math4.ai>

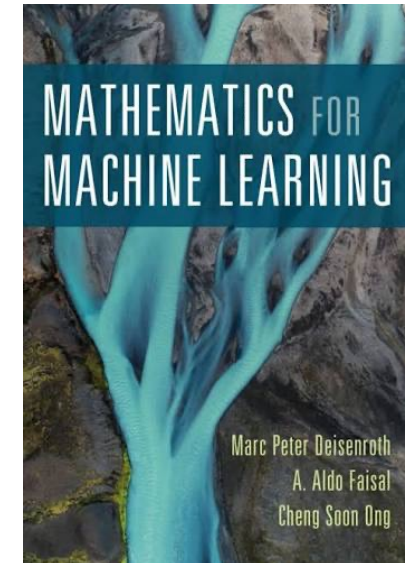
First session tomorrow!



Floor Eijkelboom

PhD Candidate @ AMLab

Math4AI Lecturer



Learning activities – Practical sessions

- You will work on the exercises from the **Practice sets**
- You also have the chance to hand in exercises from the **HW sets** to get feedback from your TA
 - You hand in the exercises on Friday
 - You get feedback on the following Wednesday/Thursday
 - **Not graded**
- You can also ask questions about the HW sheets and lab assignments

Session 1 (Wed/Thu)

Guided resolution of
exercises (2h)

Session 2 (Thu/Fri)

Guided resolution of
exercises (1h)

Q&A (1h)

Grading

Lab assignments

- Python Notebooks
- Submitted on Codegra.de (access via Canvas)
- We take the average of best 4 out of 5 grades

10%

HW-based exams

- Contain exercises based on HW sets
- Two exams (Weeks 3 and 6)
 - Week 3: HW exam 1 (15%)
 - Week 6: HW exam 2 (25%)
- Duration: 2h each

40%

Final exam

- Practice exams will be posted on Canvas
- On week 8
- Duration: 3h

50%

$$\text{final grade} = \max\{\text{FE}, 0.1 \cdot L + 0.15 \cdot \text{HE1} + 0.25 \cdot \text{HE 2} + 0.5 \cdot \text{FE}\}$$

- If you take the resit, the resit exam grade replaces the final exam grade (but the HW exams and labs can still count)
- HW grades from previous years are **not** transferrable
- More details and FAQ on Canvas

Instructions and communication

- You can find all relevant information and materials on **Canvas**
- We will send announcements through **Canvas**
 - We recommend activating email notifications!
- How to ask questions:
 - Admin questions and concrete questions about the course content and materials → **Ed discussion** (public if possible so others can see!)
 - Broad conceptual questions (e.g., "I don't understand MLE) → ask your TAs during the seminars
 - Personal matters → e.j.bekkers@uva.nl & a.monroymunoz@uva.nl or the study advisor.

Probability Theory

Probability Theory

Provides a consistent framework for the quantification and manipulation of uncertainty.

(Bishop)

In pattern recognition, uncertainty comes from:

- The noise of the measurements
- Finite size of datasets

The **probability** of an event is...

The fraction of times the event occurs in experiments

Frequentist view

Quantification of plausibility or strength of the belief of the event

Bayesian view

Events

An **event** is a subset of outcomes of a random experiment

Example: rolling a dice

- Experiment: rolling a dice and observing the outcome

- Sample space: $\Omega = \{1, 2, 3, 4, 5, 6\}$

- Examples of events:

- Getting the number 1: $A = \{1\}$ $P(A) = \frac{1}{6}$

- Getting a number greater or equal than 4: $B = \{4, 5, 6\}$ $P(B) = \frac{3}{6} = \frac{1}{2}$

- Getting an odd number: $C = \{1, 3, 5\}$ $P(C) = \frac{1}{2}$

Laplace Rule

$$\mathbb{P}[A] = \frac{\text{\# favorable cases to } A}{\text{\# possible cases}}$$

Quiz time!

A survey about rain and bike usage in Amsterdam finds that it rains on 40% of days. Overall, 60% of people cycle to work, and on rainy days this percentage decreases to 40%.



What is the probability that it rains or a person cycles to work?

a) 0.6

b) 0.76

c) 0.84

d) 0.96

R : it rains

$$P(R) = 0.4$$

C : someone cycles to work

$$P(C) = 0.6$$

$$P(C|R) = 0.4$$

$$P(C|R) = \frac{P(C \cap R)}{P(R)} \Rightarrow P(C \cap R) = 0.4 \cdot 0.4 = 0.16$$

$$\begin{aligned} P(R \cup C) &= P(R) + P(C) - P(R \cap C) \\ &= 0.4 + 0.6 - 0.16 = 0.84 \end{aligned}$$



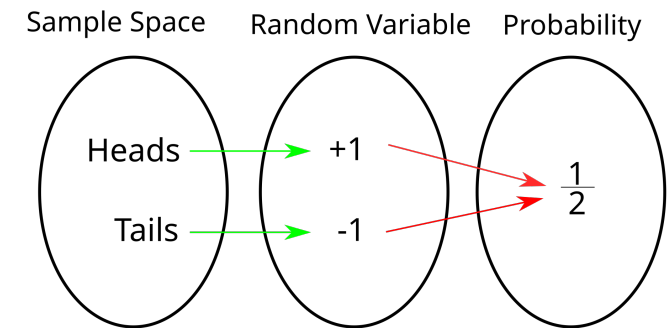
Random variables

A **random variable** is a quantity whose value is determined by the outcome of a random experiment

- Can be discrete or continuous
- Takes values in a measurable space $\mathcal{X}$
- We can characterize it by a probability distribution: $X \sim p(x)$

Examples:

- Throwing a dice: $\mathcal{X} = \{1, 2, 3, 4, 5, 6\}$
- Flipping a coin: $\mathcal{X} = \{-1, 1\}$
- Measuring height of a randomly picked ML1 student: $\mathcal{X} = \mathbb{R}^+$



Source: Wikipedia

Random variables - Probability distributions

Describe how probabilities are assigned to possible values $x \in \mathcal{X}$ of a random variable X .

Probability Density Function (PDF):

$$\mathbb{P}[a \leq X \leq b] = \int_a^b p(x) dx$$

A valid PDF p must satisfy:

$$p(x) \geq 0 \quad \forall x \in \mathcal{X} \quad \int_{\mathcal{X}} p(x) dx = 1$$

Cumulative Distribution Function (CDF):

$$P(x) = \mathbb{P}[X \leq x] = \int_{-\infty}^x p(t) dt$$

**In the discrete case,
integrals turn into sums!**

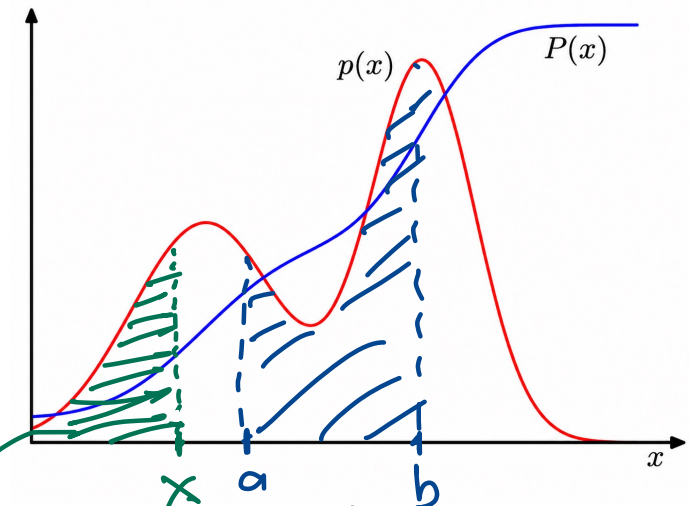


Figure: A PDF and its respective CDF
(Bishop 1.12)

Random variables – Marginal & conditional distributions

We **marginalize** using the **sum rule**:

$$p(x) = \int_y p(x, y) dy$$

(also known as Law of total probability)

Product rule:

$$p(x, y) = p(x)p(y|x)$$

Consequently, the **conditional distribution** is

$$p(y|x) = \frac{p(x, y)}{p(x)}$$

$$p(x=1, y=1) = \frac{3}{\sum p(X, Y)}$$

$$P(X=4|Y=2) = \frac{P(X=4 \cap Y=2)}{P(Y=2)}$$

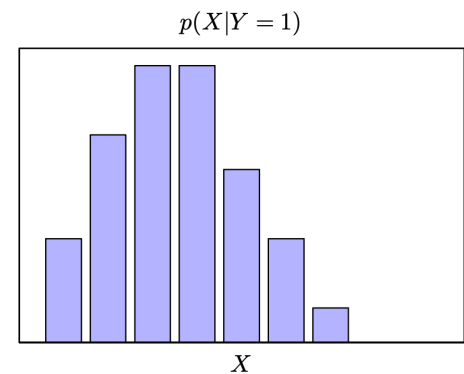
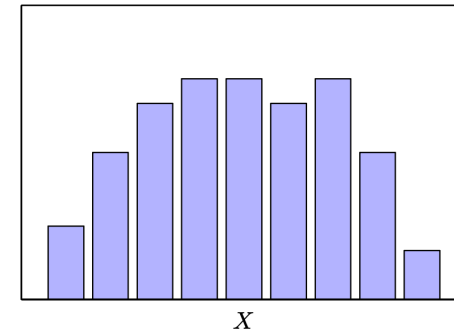
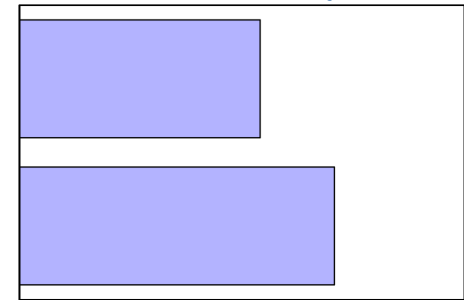
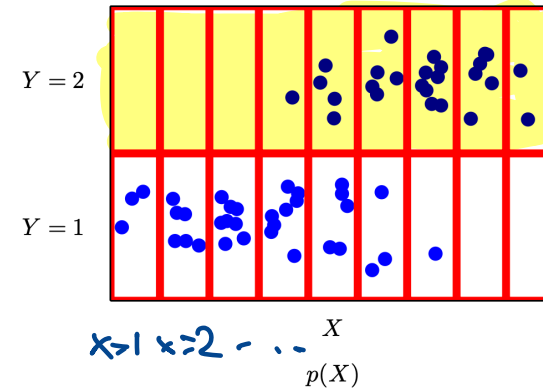


Figure: Marginal and conditional distributions for a bivariate random variable (X, Y) (Bishop 1.11)

**In the discrete case,
integrals turn into sums!**

Random variables - Independence

Two random variables X and Y are **independent** iff measuring X gives no information on Y , and vice versa.

Formally:

$$X \perp Y \Leftrightarrow p(x, y) = p(x) p(y) \\ \Leftrightarrow p(x|y) = p(x)$$

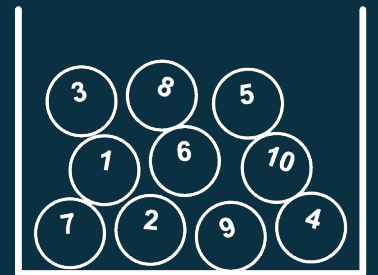
$$\forall x \in \mathcal{X}, y \in \mathcal{Y} \\ p(x|y) = \frac{p(x, y)}{p(y)} \Rightarrow \\ \Rightarrow p(x, y) = p(x|y) \cdot p(y) \\ p(x|y) \cdot \cancel{p(y)} = p(x) \cancel{p(y)}$$

Quiz time!

A box contains exactly 10 balls, numbered 1 through 10. A single ball is drawn randomly from the box. We define the following RVs:



- $X = 1$ if the number on the ball is even, and 0 otherwise.
- $Y = 1$ if the number on the ball is 1, and 0 otherwise.
- $Z = 1$ if the number on the ball is less than or equal to 4, 0 otherwise.



Which pair(s) of variables are independent?

a) ~~X, Y~~ $P(X=1, Y=1) = 0$
 $P(X=1)P(Y=1) = \frac{1}{2} \cdot \frac{1}{10} \neq 0 \quad \Rightarrow X \not\perp Y$

b) X, Z $P(X=1, Z=1) = \frac{2}{10} = \frac{1}{5}$
 $P(X=1)P(Z=1) = \frac{1}{2} \cdot \frac{4}{10} = \frac{1}{5} \quad \Rightarrow \text{All combinations of } X \in \{0, 1\} \Rightarrow X \perp Z$

c) ~~Y, Z~~

Random variables – Bayes rule

$$p(y | x) = \frac{p(x|y)p(y)}{p(x)}$$

- Allows us to update a prior probability based on new evidence
- Base of **Bayesian inference** (next lecture)

Exercise: Derive Bayes rule from the product rule

$$\begin{aligned} p(x, y) &= p(x|y)p(y) \\ p(x, y) &= p(y|x)p(x) \end{aligned} \left\{ \begin{aligned} &\Rightarrow p(x|y)p(y) = p(y|x)p(x) \\ &\Rightarrow p(y|x) = \frac{p(x|y)p(y)}{p(x)} \end{aligned} \right.$$

Random variables – Discrete vs continuous

	Discrete	Continuous
Additivity	$\mathbb{P}[X \in A] = \sum_{x \in A} p(x)$	$\mathbb{P}[X \in (a, b)] = \int_a^b p(x) dx$
Positivity	$p(x) \geq 0$	
Normalization	$\sum_x p(x) = 1$	$\int_{-\infty}^{\infty} p(x) dx = 1$
Sum Rule	$p_X(x) = \sum_{y \in \mathcal{Y}} p_{X,Y}(x, y)$	$p_X(x) = \int_{-\infty}^{\infty} p_{X,Y}(x, y) dy$
Product Rule	$p_{X,Y}(x, y) = p_{X Y}(x y) p_Y(y)$	

Exercise - Two discrete variables

Random variables:

$$X \in \{1,2,3,4,5\}, \quad Y \in \{1,2,3\}$$

N : total number of instances

n_{ij} : # instances where $X = x_i$ and $Y = y_j$

Joint PDF:

$$p(X = x_i, Y = y_j) =$$

Marginal PDF:

$$p(X = x_i) =$$

Conditional PDF:

$$p(X = x_i | Y = y_j) =$$

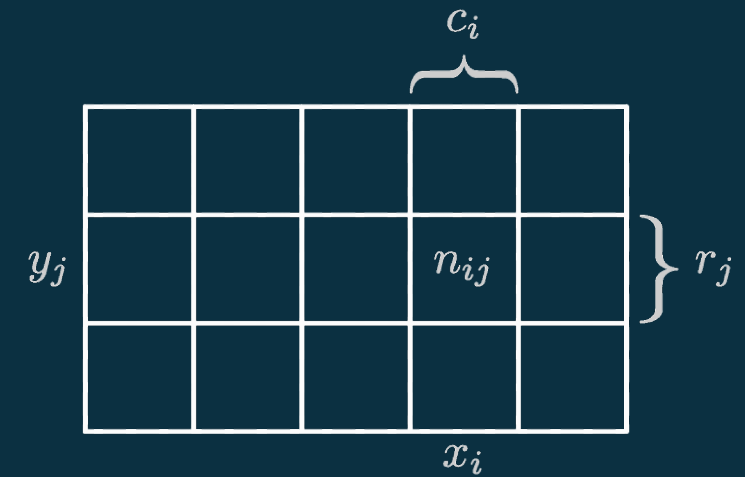


Figure: Two random variables (Bishop 1.10)

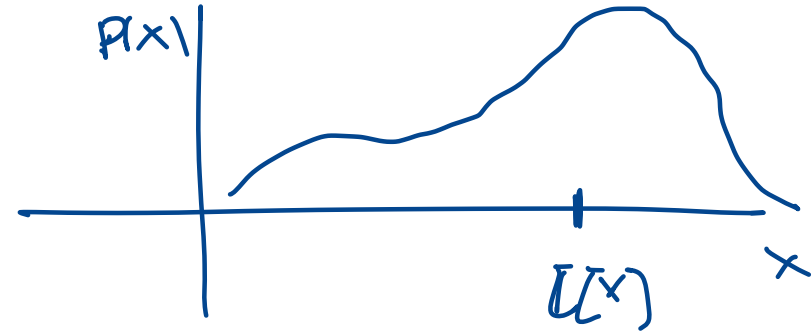
Random Variables - Moments

Expectation/Expected value:

$$\mathbb{E}[X] = \mathbb{E}_{x \sim p(x)}[X] = \int_{-\infty}^{\infty} x p(x) dx$$

More generally,

$$\mathbb{E}[f(X)] = \mathbb{E}_{x \sim p(x)}[f(X)] = \int_{-\infty}^{\infty} f(x) p(x) dx$$



The expectation is a **linear** operator. For any constants a, b :

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

$$\mathbb{E}[\text{constant}] = \text{constant}$$

Random Variables - Moments

Variance: Expected quadratic distance between X and its expected value

$$\text{var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \int (x - \mathbb{E}[x])^2 p(x) dx$$

$$\begin{aligned} &= \mathbb{E}[x^2 - 2x\mathbb{E}[x] + \mathbb{E}[x]^2] \\ &= \mathbb{E}[x^2] - 2\mathbb{E}[x\mathbb{E}[x]] + \mathbb{E}[\mathbb{E}[x]^2] \quad (\text{linearity of exp}) \\ &= \mathbb{E}[x^2] - \underbrace{2\mathbb{E}[x]\mathbb{E}[x]}_{\mathbb{E}[x]^2} + \mathbb{E}[x]^2 \\ &= \mathbb{E}[x^2] - \mathbb{E}[x]^2 \end{aligned}$$

Properties:

- $\text{var}[X] \geq 0$
- For any constant c :
 - $\text{var}[c] = 0$
 - $\text{var}[X + c] = \text{var}[X]$
 - $\text{var}[cX] = c^2 \text{var}[X]$
- $\text{var}[X + Y] = \text{var}[X] + \text{var}[Y] + 2\text{cov}[X, Y]$

Random Variables - Moments

Covariance: measures the extent to which two random variables vary together

$$\text{cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \int (x - \mathbb{E}[X])(y - \mathbb{E}[Y])p(x, y) dx dy$$

= • • (similar to previous slide)

$$= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Properties:

- $\text{cov}[X, X] = \text{var}[X]$
- $\text{cov}[X, Y] = \text{cov}[Y, X]$
- For any constants a, b :
 - $\text{cov}[X, a] = 0$
 - $\text{cov}[aX, bY] = ab\text{cov}(X, Y)$
 - $\text{cov}[X + a, Y + b] = \text{cov}(X, Y)$
- $X \perp Y \Rightarrow \text{cov}[X, Y] = 0$

Counterexample:

$$X \sim U[-1, 1]$$

$$Y = X^2$$

$$\text{cov}[X, Y] = 0$$

BUT $X \not\perp Y$



Quiz time!

We consider the random variable X with density

$$f(x) = \lambda x^{\lambda-1}, \quad 0 \leq x \leq 1$$



What is the expected value of X ?

a) $\frac{1}{\lambda+1}$

b) $\frac{\lambda}{\lambda+1}$

c) $\frac{\lambda+1}{\lambda}$

d) λ

Random Variables – Moments (multivariate case)

Given a random vector $\mathbf{X} = (X_1, X_2, \dots, X_N)$, we define:

- Expected value:

$$\mathbb{E}[\mathbf{X}] = (\mathbb{E}[X_1], \mathbb{E}[X_2], \dots, \mathbb{E}[X_N])$$

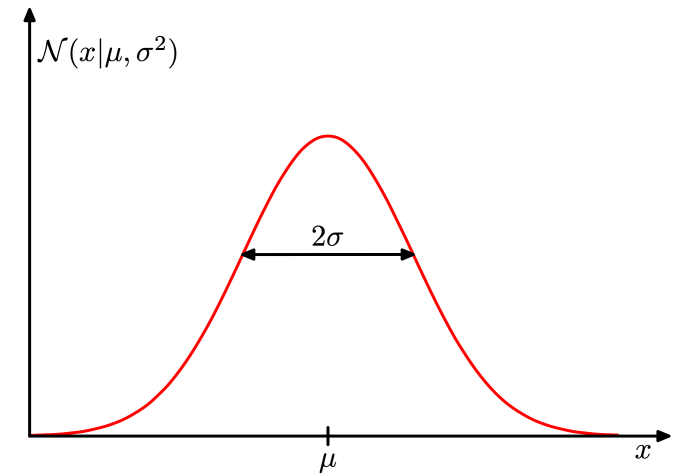
- Covariance matrix:

$$\text{Var}[\mathbf{X}] = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T] = \mathbb{E}[\mathbf{X}\mathbf{X}^T] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\mathbf{X}]^T$$

Random Variables – The Gaussian Distribution

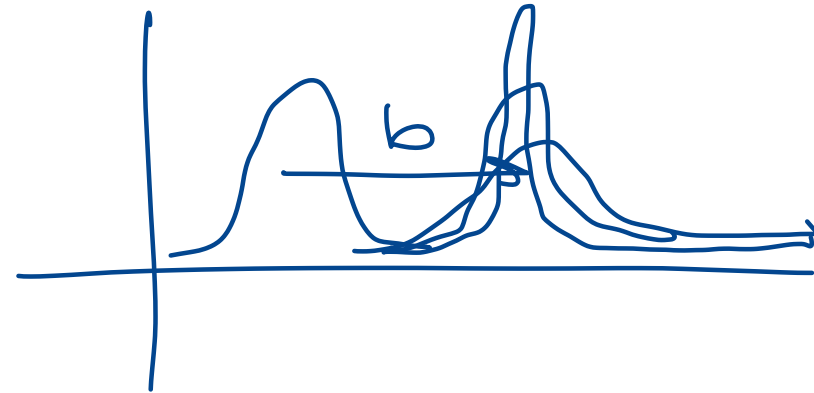
PDF:

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$



Moments: Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Then,

- $\mathbb{E}[X] = \mu$
 - $\text{var}[X] = \sigma^2$
- } **Proofs in practice set 0**



Affine transformations:

If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = aX + b$, where $a, b \in \mathbb{R}$ and $a \neq 0$, then

$$Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2).$$

Random Variables – The Gaussian Distribution

Multivariate case:

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right).$$

$\downarrow$ dim.

Moments: Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Then,

- $\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu}$
- $\text{cov}[\mathbf{X}] = \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1d} \\ \sigma_{21} & \sigma_2^2 & & \vdots \\ \vdots & & \ddots & \\ \sigma_{d1} & \dots & & \sigma_d^2 \end{pmatrix}$ where $\begin{cases} \sigma_i^2 = \text{var}[x_i] \\ \sigma_{ij} = \text{cov}[x_i, x_j] \end{cases}$

Affine transformations:

If $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where $\mathbf{A} \in \mathbb{R}^{m \times d}$, $\mathbf{b} \in \mathbb{R}^m$, then

$$\mathbf{Y} \sim \mathcal{N}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

Random Variables - The Gaussian Distribution

Let's show the calculation for the mean. Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\begin{aligned}\mathbb{E}[\mathbf{X}] &= \underline{\mathbf{A}} \mathbb{E}[\underline{\mathbf{z}}] + \underline{\boldsymbol{\mu}} \quad \left(\begin{array}{c} \text{Linearity} \\ \text{of } \mathbb{E} \end{array} \right) \\ &= \underline{\mathbf{A}} \underline{\mathbf{0}} + \underline{\boldsymbol{\mu}} = \underline{\boldsymbol{\mu}}\end{aligned}$$

$\uparrow$

Let $\underline{\mathbf{x}} = \underline{\mathbf{A}} \underline{\mathbf{z}} + \underline{\boldsymbol{\mu}}$, where
 $\underline{\mathbf{z}} \sim \mathcal{N}(\underline{\mathbf{0}}, \underline{\mathbf{I}})$
and
 $\underline{\mathbf{A}} \underline{\mathbf{A}}^T = \boldsymbol{\Sigma}$

$$\mathbb{E}[\underline{\mathbf{z}}] = \int_{\mathbb{R}^d} \underbrace{\underline{\mathbf{z}}}_{\text{odd}} \underbrace{p(\underline{\mathbf{z}})}_{\text{even}} d\underline{\mathbf{z}} = \underline{\mathbf{0}}$$

symmetric domain $\rightarrow$

$$p(\underline{\mathbf{z}}) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2} \underline{\mathbf{z}}^T \underline{\mathbf{z}}\right)$$

$$p(\underline{\mathbf{z}}) = p(-\underline{\mathbf{z}}) \Rightarrow p(\underline{\mathbf{z}}) \text{ is an even function}$$

Random Variables – ML Notation

If you learned probability in a non-ML setting, you might be used to the following notation (which is the one we have followed up to this slide):

- Capital letters for random variables:
 X : height of a ML1 student
- Lowercase for instances of the variable
 $x = 175 \text{ cm}$

But in most ML literature we lose this distinction and we capitalize based on the type of mathematical object that we're working with:

Scalar-valued RV: x

Vector-valued RV: $\mathbf{x}$

Matrix-valued RV: $\mathbf{X}$

Tensor-valued RV: $\mathbf{X}$

And this is what we (and Bishop) will do from now on

In practice: people may use different notations, so better to learn to deduce by context!

That was it!

Tomorrow: Math4AI – Vector Calculus

Wednesday:

- Second Lecture
- First tutorial (take a look at the exercise sheet in advance)

HW set 1 is out

Lab 1 is out – Deadline next Monday

Next Lecture: Estimation

- **Estimation and Bias**
- **Maximum Likelihood Estimation**
- **Bayesian Estimation**